

Lab 7

(cont...)

Goodness-of-fit for linear regression

“All models are wrong, but some are useful.”

George Box (1919-2013), pioneering statistician

How well does a model fit our data? In many cases, answers are based on analysis of the residuals:

- normality of residuals;
- homoscedasticity: constant residual variance throughout the range of the predicted values.

Another measure is that of “variance explained” R^2 (see C11 slide 14).

```
# example adapted from [1] p. 157
# house_sales.csv taken from
# https://drive.google.com/drive/folders/0B98gpkK5EJemYnJ1ajA1ZVJwMzg

house <- read.csv(file.path('C:\\Users\\Marina\\Desktop\\curs Statistics for Data
Science\\house_sales.csv'), sep='\\t')
#house <- read.csv(file.path('C:\\Users\\Marina\\Desktop\\curs Statistics for Data
#Science\\house_sales1.csv'))

house_98105 <- house[house$ZipCode == 98105, ] #select the houses in an
#area

sample_house_98105<- house_98105[sample(nrow(house_98105),30), ]

model <- lm(AdjSalePrice ~ SqFtTotLiving + SqFtLot + Bathrooms +
            Bedrooms + BldgGrade, data=sample_house_98105)

summary(model)

#Multiple R-squared: 0.8218
#that is  $R^2$  -- we say that 82.18% of variability can be explained through the
#linear model – by this measure, the model fits well

shapiro.test(residuals(model))
```

Shapiro-Wilk normality test

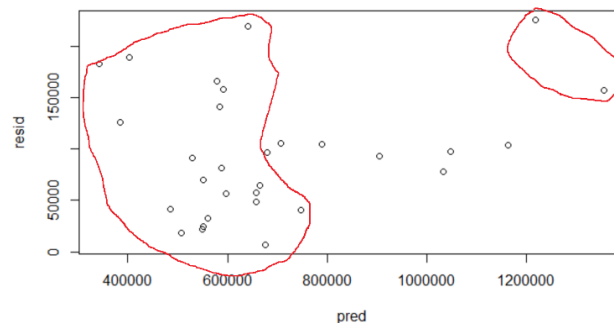
```
data: residuals(model)
W = 0.96923, p-value = 0.5184
#normality assumption is met for our model
#Obs. Models are robust to non-normality to a certain extent
```

homoscedasticity

```
resid <- abs(residuals(model)) #interested in the magnitude of the residuals
#names(resid)<-NULL           #to remove the names of the resid vector
```

```
pred<-predict(model)
```

```
plot(pred,resid)
```



We see a greater variation of residuals for lower-valued homes (left) and for the higher-valued homes (right) and much less variation for the mid-valued homes -- homoscedasticity assumption not met. It may indicate a “cause” missing from the linear equation (causes ->effect) -- worth investigating other (linear) models.

Also check the Breusch–Pagan test against heteroskedasticity.

Repeat the same steps for the goodness-of-fit for the example below (taken from Lab6):

```
longley
X <- longley[, "Employed"] # number of people employed
Y <- longley[, "Population"] # population ≥ 14 years of age

model1<-lm(X~Y)
```

What is your conclusion after this analysis?

Nonlinear regression

!!Please read C12

```
# we use house_sales.csv to generate the data for the nonlinear regression
#examples

house <- read.csv(file.path('C:\\Users\\Marina\\Desktop\\curs Statistics for Data S
cience\\house_sales.csv'), sep='t')

house_98105 <- house[house$ZipCode == 98105, ]

sample_house_98105<- house_98105[sample(nrow(house_98105),30), ]
#random selection of 30 rows from the dataset for a clearer visualization of the
#points

lm_30 <- lm(AdjSalePrice ~ SqFtTotLiving + SqFtLot + Bathrooms +
            Bedrooms + BldgGrade, data=sample_house_98105)
resid <- abs(residuals(lm_30))
pred<-predict(lm_30)

#-----
#we consider (pred,resid) as (cause,effect) data for nonlinear regression model

plot(pred,resid, ylim = c(-100000,300000))

library(splines)

#interpolation using splines
func = splinefun(x=pred, y=resid, method="natural",ties=mean)

#pred are the knots
#method= the type of the spline used
#ties = the name of a function specifying how to handle duplicate x values. The y
#values corresponding to the same x value are passed to the function, which
#return a single number (mean in our case)

xmin=min(pred)
xmax=max(pred) # [xmin,xmax] is the interval for regression

new_points=seq(xmin,xmax,length=50)

allpoints=c(new_points, pred) #include the knots among the points for prediction
allpoints_ordered = allpoints [order(allpoints)] #necessary to visualize well the
#regression curve
```

```
#lines(pred,func(pred), type="p", col=1)
```

```
lines(allpoints_ordered,func(allpoints_ordered), type="l", col=2)
```

```
#smoothing using splines
```

```
fit <- smooth.spline(pred, resid, all.knots=T, spar=0.7)
```

```
#try for different values of spar in [0,1]
```

```
#spar is alpha parameter in SS(g) – see C12 slide 3
```

```
#if alpha=0 the regression curve goes through all the knots – like interpolation
```

```
#if alpha=1 the regression curve becomes smoother and it doesn't go through
```

```
all the knots
```

```
res <- stats::predict.smooth.spline(fit, allpoints_ordered)$y
```

```
lines(allpoints_ordered,res, type="l", col=4) # plot the fitted spline
```